

#### 4.4.4 Stationary waves

| Learning outcomes                                                                                | Additional guidance |
|--------------------------------------------------------------------------------------------------|---------------------|
| <i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>    |                     |
| (a) stationary (standing) waves using microwaves, stretched strings and air columns              |                     |
| (b) graphical representations of a stationary wave                                               |                     |
| (c) similarities and the differences between stationary and progressive waves                    |                     |
| (d) nodes and antinodes                                                                          |                     |
| (e) (i) stationary wave patterns for a stretched string and air columns in closed and open tubes |                     |

- (ii) techniques and procedures used to determine the speed of sound in air by formation of stationary waves in a resonance tube
- (f) the idea that the separation between adjacent nodes (or antinodes) is equal to  $\lambda/2$ , where  $\lambda$  is the wavelength of the progressive wave
- (g) fundamental mode of vibration (1st harmonic); harmonics.